

Sahi Dawa: AI Prescription Transparency & Care Assistant
Pak Angels GenAI Hackathon, Cohort 11 (Health Care Category)

Team: Abdullah Aamir (Lead), Abdullah, Faraz Ahmed Memon, Esha Inam, Absar Ahmed, Afra Naz

(Final locked proposal pre-PRD)

Following everyone's feedback, here is our final locked concept and MVP scope before we move into the PRD and start building.

The Problem

Patients often don't know whether the medicines prescribed to them are:

- Appropriate for their diagnosed condition.
- Available as cheaper generic alternatives.
- More expensive than necessary because of brand selection.
- Being repeatedly prescribed unnecessarily, particularly antibiotics.

Research from Pakistan has documented concerns around pharmaceutical incentives [1], irrational antibiotic prescribing and resulting antimicrobial resistance [2,3,4], and significant brand-vs-generic cost gaps that affect patients' ability to afford treatment [5,6,7]. **We are not accusing individual doctors.** Sahi Dawa exists to give patients transparency and cost awareness so they can discuss treatment more confidently with qualified healthcare professionals.

Our Positioning

"We don't replace the doctor. We make the prescription transparent."

Sahi Dawa is a GenAI-powered prescription transparency assistant that uses verified Pakistani medicine data and patient history to help people understand their prescriptions, compare generic/cost information, and prepare better questions for their healthcare professionals. It does **not** independently prescribe, discontinue, or substitute medication.

How It Works (User Journey)

1. Patient enters diagnosis + medicine name + dosage.
2. System identifies the medicine and retrieves verified catalogue data.
3. AI explains the medicine using retrieval-based reasoning (RAG), not free-form guessing.
4. System compares brand vs. same-active-ingredient alternatives and cost.
5. Patient-friendly report generated with flags worth discussing.
6. Result saved to patient history.
7. System detects patterns across visits (e.g., repeated antibiotics).
8. Shareable Patient Health Summary generated.

Final MVP Scope

Feature	Status
Prescription input (diagnosis, medicine, dosage)	✅ MVP
Medicine identification	✅ MVP
Active ingredient extraction	✅ MVP
Verified medicine catalogue (DRAP-referenced)	✅ MVP
RAG-based medicine explanation	✅ MVP
Same-active-ingredient brand comparison	✅ MVP
Cost/price transparency	✅ MVP
Patient history (agentic memory)	✅ MVP
Historical pattern detection	✅ MVP
Patient Health Summary (shareable report)	✅ MVP
Autonomous medicine replacement	❌ Never
Automated DRAP updates	🕒 Future
Handwriting/OCR input	🕒 Future

What We Will NOT Build for This Hackathon

Automated DRAP scraping · full Pakistani medicine database · live pharmacy marketplace · autonomous prescribing/substitution · AI diagnosis · full electronic medical record · complex OCR · diet/lifestyle engine · vitals-based treatment recommendations

Data Strategy

- Reference source: DRAP (Drug Regulatory Authority of Pakistan) — used as reference, not live integration
- Hackathon approach: manually curated catalogue of ~30–50 medicine records
- Schema: `medicine_id`, `brand_name`, `generic_name`, `active_ingredient`, `strength`, `dosage_form`, `pack_size`, `registration_number`, `manufacturer`, `price`, `price_effective_date`, `drap_source_url`, `last_verified`, `data_status`
- Real vs. synthetic: use verified real data where possible; clearly label anything else as **DEMO/SYNTHETIC**. **Never present fabricated data as real DRAP data.**
- If a medicine isn't in our catalogue, the system says so — it never guesses.

Safety Guardrails

The AI must never invent prices, ingredients, strengths, registration info, indications, or alternatives. Clinical language stays in the form of *"here's what our verified data shows, discuss with your doctor"* — never *"this is wrong for you"* or a directive to switch/stop medication.

Why This Is Genuinely Agentic

Not agentic just because it has a chatbot. The loop is:

Retrieve → **Analyze** → **Remember** → **Compare with previous encounters** → **Identify pattern** → **Inform patient**

Example: Visit 1 antibiotic, Visit 2 antibiotic, Visit 3 antibiotic → *"Your previous records show repeated antibiotic prescriptions. Please discuss whether continued antibiotic use is clinically necessary with your healthcare professional."*

Team Roles (open, choose based on interest)

AI/LLM · Pharmacy & medicine validation · Backend/API · Database & agentic memory · Frontend/UI · Dataset creation · Testing · Research · Presentation/pitch

References

On pharmaceutical incentives & prescribing behaviour

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On brand vs. generic pricing and cost burden

5. Barriers and Facilitators of Generic Medication Substitution Among Healthcare Providers in Pakistan: A Qualitative Study. *ScienceDirect*, 2026
6. Brand versus generic dispensing trend for ciprofloxacin, levofloxacin, and moxifloxacin among pharmacies of Karachi, Pakistan. *PMC*, 2017.
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